Recap: Lecture 29

In Lecture 29, we

- Introduced a FT table
- Introduced convolution and shift theorems for FT

$$\mathscr{F}\{f * g\} = \widehat{f}(k)\,\widehat{g}(k), \quad \mathscr{F}\{f(x-y)\} = \mathrm{e}^{-iky}\,\widehat{f}(k) \;\; (y \in \mathbb{R}),$$

$$\mathscr{F}\{\mathrm{e}^{iqx} f\} = \widehat{f}(k-q) \;\; (q \in \mathbb{R})$$

- Learnt rules to help apply FTs to ODEs

$$\mathscr{F}\left(f^{(n)}\right) = (ik)^n \widehat{f}, \quad \mathscr{F}(x^n f) = i^n \frac{\mathrm{d}^n \widehat{f}}{\mathrm{d}k^n}$$

Lecture 30

In Lecture 30, we will:

1. Apply the FT to the Airy ODE
2. Apply the FT to a PDE (1D heat equation)

Exercise 6.9 Revisited

$$y''(x) - y(x) = \delta(x) \quad \longrightarrow \quad \mathrm{FT} \quad \longrightarrow \quad y(x) = \frac{-1}{2}\mathrm{e}^{-|x|}$$

Exercise 6.9 Revisited

$$y''(x) - y(x) = \delta(x) \quad \longrightarrow \quad \mathrm{FT} \quad \longrightarrow \quad y(x) = \frac{-1}{2}\mathrm{e}^{-|x|}$$

$$y''(x) - y(x) = 0 \quad \Rightarrow \quad y(x) =$$

$\lambda^2 - 1 = 0 \Rightarrow$

Exercise 6.9 Revisited

$$y''(x) - y(x) = \delta(x) \quad \longrightarrow \quad \text{FT} \quad \longrightarrow \quad y(x) = \frac{-1}{2}\mathrm{e}^{-|x|}$$

$$y''(x) - y(x) = 0 \quad \Rightarrow \quad y(x) = C_+\, y_+(x) + C_-\, y_-(x) \quad \text{where} \quad y_\pm(x) = \mathrm{e}^{\pm x}$$

Exercise 6.9 Revisited

$$y''(x) - y(x) = \delta(x) \quad \longrightarrow \quad \text{FT} \quad \longrightarrow \quad y(x) = \frac{-1}{2}\mathrm{e}^{-|x|}$$

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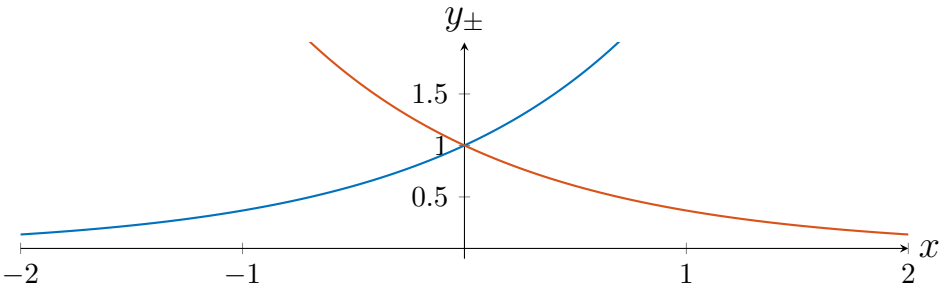
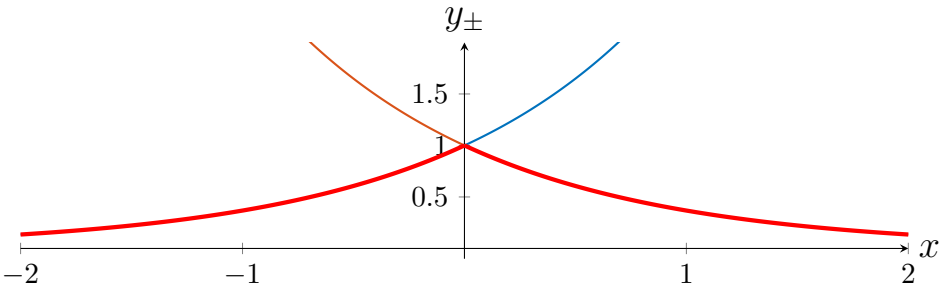


Exercise 6.9 Revisited

$$y''(x) - y(x) = -2\,\delta(x) \quad \longrightarrow \quad \text{FT} \quad \longrightarrow \quad y(x) = \mathrm{e}^{-|x|}$$

$$y''(x) - y(x) = 0 \quad \Rightarrow \quad y(x) = C_+\, y_+(x) + C_-\, y_-(x) \quad \text{where} \quad y_\pm(x) = \mathrm{e}^{\pm x}$$



Recap (Topic 2): Airy's equation $y''(x) - x\,y(x) = 0$

$$\boxed{\text{Has no solutions in terms of elementary functions}}$$

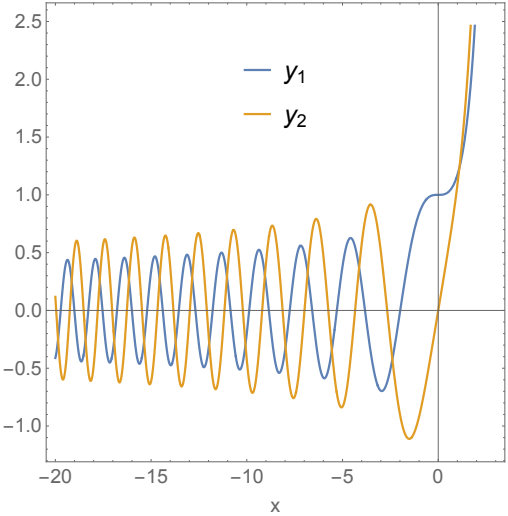
Two independent series solutions:

$$y_1(x) = 1 + \sum_{k=1}^{\infty} \frac{x^{3k}}{2 \cdot 3 \cdot 5 \cdot 6 \cdots (3k-1) \cdot 3k}$$

$$y_2(x) = x + \sum_{k=1}^{\infty} \frac{x^{3k+1}}{3 \cdot 4 \cdot 6 \cdot 7 \cdots 3k \cdot (3k+1)}$$

The general solution of Airy's eqn.

$$y(x) = C_1\, y_1(x) + C_2\, y_2(x)$$



Recap (Topic 2): Airy functions

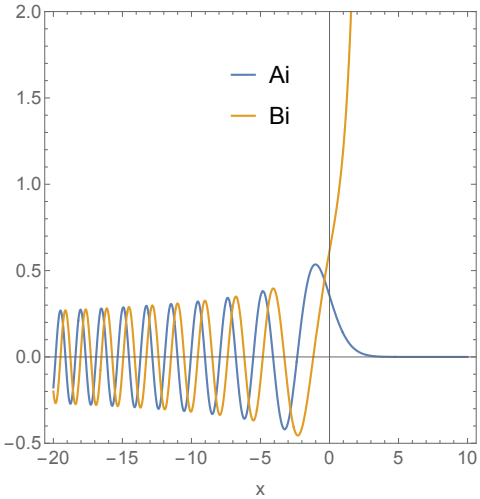
The *Airy functions* are

$$\mathrm{Ai}(x) = \alpha\left[y_1(x) - \beta y_2(x)\right]$$

$$\mathrm{Bi}(x) = \sqrt{3}\alpha\left[y_1(x) + \beta y_2(x)\right]$$

where $\alpha \approx 0.355$ and $\beta \approx 0.729$.

Note Ai(x) $\to 0$ as $x \to \infty$



Exercise 6.10: Use FTs to solve Airy's equation $y''(x) - xy(x) = 0$

Apply FT :

$$\underbrace{\mathcal{F}\{y'' - xy\}}_{\mathcal{F}\{y''\} - \mathcal{F}\{xy\}} = \mathcal{F}\{0\} = 0$$

$$-k^2\hat{y} - i\hat{y}' = 0$$

ODE in k-space

$$\hat{y}' - ik^2\hat{y} = 0$$

IF : $I = e^{\int -ik^2\,dk} = e^{-ik^3/3}$

$$\therefore (I\hat{y})' = 0 \Rightarrow I\hat{y} = C$$

$$\Rightarrow \hat{y} = C/I = Ce^{ik^3/3}$$

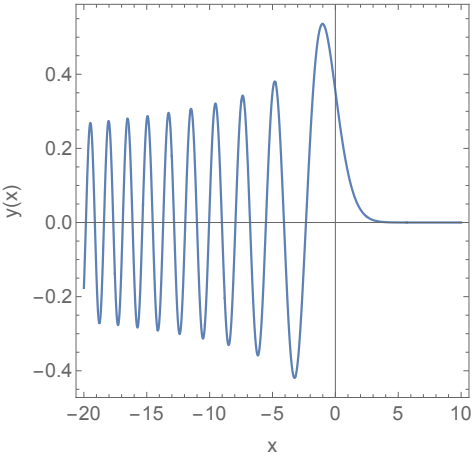
$$\therefore \quad y(x) = F^{-1}\{\hat{y}\}$$

$$= F^{-1}\{Ce^{ik^3/3}\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} C e^{ik^3/3} e^{ikx}\, dk$$

$$= \frac{C}{2\pi}\int_{-\infty}^{\infty} e^{i(k^3/3 + kx)}\, dk$$

We have found Ai in integral form

$$y(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{i}\,(k^3/3 + k\,x)}\ \mathrm{d}k$$



Transforms applied to PDEs

- We've already seen that Laplace transforms and Fourier transforms turn ODEs into algebraic equations
- Transforms can also be used to turn PDEs into ODEs

Transforms applied to PDEs

- We've already seen that Laplace transforms and Fourier transforms turn ODEs into algebraic equations
- Transforms can also be used to turn PDEs into ODEs
- Let $u(x,t)$ be defined for $x \in \mathbb{R}$ and $t > 0$.
  - Let $\widehat{u}(k,t) = \mathscr{F}\{u(x,t)\}$
  - $\mathscr{F}\left(\dfrac{\partial^n u}{\partial x^n}\right) = (ik)^n\,\widehat{u}(k,t)$, etc.

Exercise 6.11: 1D Heat Equation

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} \quad \text{where} \quad \alpha > 0, \quad x \in \mathbb{R}, \quad t > 0,$$

subject to the initial condition $u(x,0) = f(x)$ and $u(x,t) \to 0$ for $x \to \pm\infty$.

Define $\hat{u}(k,t) = F\{u\}$

Apply FT (wrt x) to PDE (1d heat eqn)

LHS : $F\{u_t\} = \hat{u}_t(k,t)$

RHS : $F\{\alpha u_{xx}\} = \alpha F\{u_{xx}\} = -\alpha k^2 \hat{u}(k,t)$

$\therefore \quad \hat{u}_t = -\alpha k^2 \hat{u}$ (ODE)

$\Rightarrow \quad \hat{u} = C(k) e^{-\alpha k^2 t}$

Need an IC : Apply FT to $u(x,0) = f(x)$

$$\Rightarrow \hat{u}(k,0) = \hat{f}(k) = F\{f\}$$

$$\therefore \quad \hat{u}(k,0) = \boxed{C(k) = \hat{f}(k)} \quad \Rightarrow \hat{u}(k,t) = \hat{f}(k)e^{-\alpha k^2 t}$$

$$u(x,t) = F^{-1}\{\hat{u}(k,t)\}$$

$$= F^{-1}\{\hat{f}(k)e^{-\alpha k^2 t}\}$$

$$= \underbrace{F^{-1}\{\hat{f}\}}_{f(x)} * F^{-1}\{e^{-\alpha k^2 t}\}$$

|| convolution thm

FT table: $F^{-1}\left\{\sqrt{\frac{\pi}{a}}\, e^{-k^2/4a}\right\} = e^{-ax^2} \quad (a>0)$

|| We need $\alpha t = 1/4a$

$\Rightarrow a = 1/4\alpha t$

$$F^{-1}\left\{\sqrt{\pi 4\alpha t}\; e^{-\alpha k^2 t}\right\} = e^{-x^2/4\alpha t}$$

$$F^{-1}\left\{e^{-\alpha k^2 t}\right\} = \frac{1}{\sqrt{4\alpha\pi t}}\, e^{-x^2/4\alpha t}$$

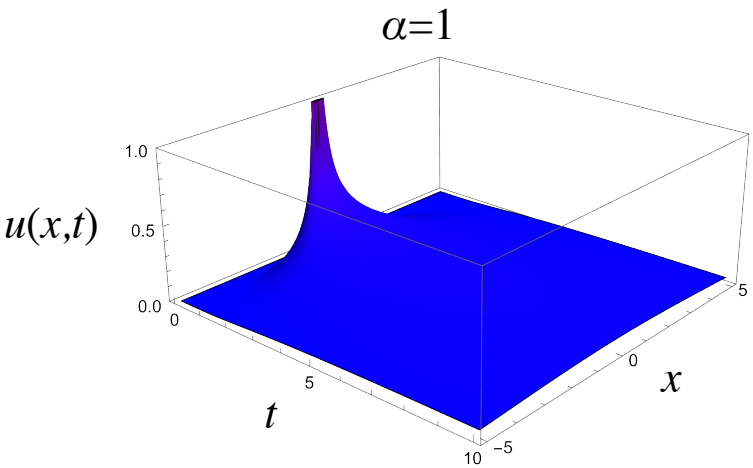
$$\therefore\; u(x,t) = f(x) * \frac{1}{\sqrt{4\alpha\pi t}}\, e^{-x^2/4\alpha t}$$

$$= \frac{1}{\sqrt{4\pi\alpha t}} \int_{-\infty}^{\infty} f(x-\xi)\, e^{-\xi^2/4\alpha t}\, d\xi$$

Find the particular solution for $f(x) = \delta(x)$.

$$u(x,t) = \frac{1}{\sqrt{4\pi\alpha t}} \int_{-\infty}^{\infty} \delta(x-\xi)\, e^{-\xi^2/4\alpha t}\, d\xi$$

$$= \frac{1}{\sqrt{4\pi\alpha t}}\, e^{-x^2/4\alpha t}$$


α=1
u(x,t)
1.0
0.5
0.0
0
5
10
t
−5
0
5
x